

Thus, the young optimally hold no equities (they actually want to short equities if they can), and equity portfolio shares increase with age.²⁹

Lynch and Tan make the point that although the correlation of labor income and equities is close to zero *unconditionally*, it can mask pronounced *conditional* correlation over the business cycle. Put simply, while there isn't much correlation between labor income and equities overall, there is significant correlation when it really matters, like when stocks crash. In their estimates, conditional shocks to permanent components of labor income are 48% correlated with conditional stocks to stock returns even though the unconditional correlation between labor income and stock returns is just 2% in their sample. Since labor income is conditionally stock-like, the young optimally hold low equity portfolio weights.

3.6. OTHER CONSIDERATIONS

The surprising preference for more equities in old age is predicted by the life-cycle model if labor income is risky, like a stock. (I reiterate: this is the opposite of standard financial planner advice.) Other circumstances, however, may warrant reducing equities as retirement approaches. Perhaps the most important is . . .

Health-Care Risk

Health-care shocks can be truly debilitating. Health care is expensive and spending is concentrated in late life. Marshall, McGarry, and Skinner (2010) report that the median medical spending in the last year of an American's life exceeds \$5,000, but for the top 5% it exceeds \$50,000. Both the mean and variance of medical expenditure rise with age, especially in old age. Health-care shocks do not mean just a one-off medical expenditure. One serious medical problem is too often followed by others. De Nardi, French, and Jones (2010) unpack health-care spending into persistent and transitory components and show that the persistent components have autocorrelations of more than 90% per year.

Health-care risk is not just about being able to pay medical bills. Serious ill health means the loss of current income for a working family and the loss of future income as well. Substantial declines in wealth and future labor income follow bad health-care shocks.³⁰ Sickness often produces a downward spiral: ill health affects earnings, and lower earnings lead to poorer health.

Some health-care costs are borne by insurance and some by the government (Medicare and Medicaid). But a lot of health-care expenses in the United States are paid by individuals. Munnell et al. (2008) estimate the present value of out-of-pocket health-care costs from age sixty-five at \$100,000 per person, excluding long-term care. Webb and Zhivan's (2010) estimates are even higher,

²⁹ They are actually slightly hump-shaped, with the old slightly decreasing their equity portfolio weights compared to the middle-aged.

³⁰ The large literature is summarized by Poterba, Venti, and Wise (2011).